

Data Cleaning Report

- **Dataset Name:** synthetic_ecommerce_sales_2025
- **File Type:** CSV (Comma Separated Values)
- **Rows:** 100000 rows
- **Columns:** 13 columns
- **Purpose:** To analyze product sales, quantity, revenue, and customer purchase patterns.

The raw dataset contains columns such as order_id, customer_id, product_category, product_price, quantity, order_date, region, payment_method, delivery_days, is_returned, customer_rating, discount_percent, revenue, month and month_year.

Issues Identified in Raw Data

- Incorrect data types – order_date stored as text instead of date format.
- Outliers - Revenue column had extreme values.

Correction of identified issues

- Order_date was converted into datetime format for proper time-based analysis.
- Revenue outliers were checked using Z-score & IQR method.
 - ✓ revenue ranges from about 4.26 to 2699.14
 - ✓ Using IQR, about 903 rows are flagged as revenue outliers
 - ✓ Using Z-score > 3, about 247 rows are flagged
 - ✓ These “outliers” mostly happen when:
 - ✓ quantity is 5 or 6
 - ✓ product_price is high
 - ✓ discount_percent is 0 or low

Important conclusion

- ✓ These are not serious dirty-data outliers in the usual sense. They are mostly valid high-revenue transactions, caused by:
 - higher quantity,
 - higher price,
 - lower discount.

❖ Reason to keep them as it is was:

- ✓ Because they are logically valid and linked to quantity and price.
- ✓ Removing them may destroy real sales behavior.

Addition of New Columns

- **month** column was created from `order_date` to support monthly sales trend analysis.
- **month_year** column was created from `order_date` to support yearly / monthly_yearly sales trend analysis.

Importance Assumptions during data cleaning

- High revenue values were not removed unless clearly unrealistic.
- Some outliers may still remain because they may represent valid business transactions.

Key Insights

- Revenue distribution is right-skewed
- Some high revenue values are due to bulk quantity sales
- Electronics category had the highest number of transactions
- Most errors were found in date formatting

Final Cleaned Dataset Summary

- After cleaning, the dataset contains 100000 rows and 15 columns.
- Missing values and duplicate rows were handled successfully.
- All columns now have correct data types and the dataset is ready for further analysis and visualization.